

Tight security of XMSS (Reduction-Based Approach)

Work in Progress

Abstract

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1 Introduction

WIP adaptation of [1]. Differences are:

1. Keep using Multi-Target Collision Resistance (SM-TCR) and Multi-Target Collision Resistance with Random Sampling (SM-rTCR), but replace Multi-Target Preimage Resistance (SM-PRE) and Multi-Target Undetectability (SM-UD) with a new (non-standard) property: Multi-Target Chain Preimage Resistance (SM-CHAIN).
2. Multi-user scenario.

The idea behind 1. is to avoid the $L \cdot v \cdot 2^w$ factor in the final security bound. This overhead comes from the event in which an adversary is able to find a direct preimage in a "hash chain" at a position just before the "earlier" revealed value. In [1], the reduction must guess which chain and which position the adversary will target, leading to the multiplicative loss. By capturing this attack scenario directly in the SM-CHAIN game, we can obtain a tighter reduction that avoids this guessing overhead. It remains to prove a bound for SM-CHAIN in the ROM (easy) and QROM (hard).

1.1 Tweakable Hash Functions

Definition 1.1 (Tweakable Hash Function). Consider sets $\mathcal{H}$ (the hash space), $\mathcal{P}$ (the public parameters space), $\mathcal{T}$ (the tweak space), and let $\mathcal{M}$ (the message space). A tweakable hash function is an efficiently computable function

$$\text{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}.$$

1.2 Multi-user properties

1.2.1 Multi-User Multi-Target Collision Resistance

Definition 1.2 (Multi-User Multi-Target Collision Resistance). Let $\text{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}$ be a tweakable hash function. Let $N \in \mathbb{N}$ be the number of users, let $p \in \mathbb{N}$ be a per-user query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\text{MULTI-SM-TCR}_{\text{Th}, N, p}(\mathcal{A})$:

Setup:

1. For each $i \in [N]$, sample $P_i \xleftarrow{\$} \mathcal{P}$ conditioned on $P_i \neq P_j$ for all $j < i$.
2. For each $i \in [N]$, initialize $Q_i := \emptyset$.

Query Phase: Run $\mathcal{A}$ with (classical) access to the following oracle:

- Hash Oracle $\mathcal{O}_{\text{target}}(i, T, M)$ where $i \in [N]$, $T \in \mathcal{T}$, $M \in \mathcal{M}$:
 - (a) If $|Q_i| \geq p$ or there exists $M' \in \mathcal{M}$ with $(T, M') \in Q_i$, return $\perp$.
 - (b) Insert (T, M) into Q_i .
 - (c) Return $\text{Th}(P_i, T, M)$.

Reveal Phase: When $\mathcal{A}$ signals to proceed:

1. Disable $\mathcal{O}_{\text{target}}$ and provide $(P_1, \dots, P_N)$ to $\mathcal{A}$.

Output: $\mathcal{A}$ outputs (i^*, j^*, M^*) with $i^* \in [N]$, $j^* \in [|Q_{i^*}|]$, and $M^* \in \mathcal{M}$.

Winning Condition: Let (T_{j^*}, M_{j^*}) be the j^* th entry in Q_{i^*} . Output 1 if and only if:

- (i) $\text{Th}(P_{i^*}, T_{j^*}, M_{j^*}) = \text{Th}(P_{i^*}, T_{j^*}, M^*)$, and
- (ii) $M^* \neq M_{j^*}$.

The advantage is:

$$\text{Adv}_{\text{Th}, N, p}^{\text{MULTI-SM-TCR}}(\mathcal{A}) := \Pr[\text{MULTI-SM-TCR}_{\text{Th}, N, p}(\mathcal{A}) \Rightarrow 1].$$

Intuition 1.3. An attacker plays a game where they first *commit* to a collection of messages—without knowing the hash function’s secret parameters—and then later attempt to find a *different* message that collides with one of their committed messages.

1.2.2 Multi-User Multi-Target Collision Resistance with Random Sampling

Definition 1.4 (Multi-User Multi-Target Collision Resistance with Random Sampling). Let $K \in \mathbb{N}$ be an integer. Let $\text{Th} : \mathcal{P} \times \mathcal{T} \times (\mathcal{M} \times \mathcal{R}) \rightarrow \mathcal{H}$ be a tweakable hash function, where the input is split into a message part $M \in \mathcal{M}$ and a randomness part $\rho \in \mathcal{R}$. Let $\text{Prop} : \mathcal{H} \rightarrow \{0, 1\}$ be a predicate on the output space. Let $N \in \mathbb{N}$ be the number of users, let $p \in \mathbb{N}$ be a per-user query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\text{MULTI-SM-rTCR}_{\text{Th}, N, p, \text{Prop}}^K(\mathcal{A})$:

Setup:

1. For each $i \in [N]$, sample $P_i \xleftarrow{\$} \mathcal{P}$ conditioned on $P_i \neq P_j$ for all $j < i$.
2. For each $i \in [N]$, initialize $Q_i := \emptyset$.

Query Phase: Run $\mathcal{A}$ with input $(P_1, \dots, P_N)$ and (classical) access to the following oracle:

- Sampling Oracle $\mathcal{O}_{\text{sample}}(i, T, M)$ where $i \in [N]$, $T \in \mathcal{T}$, $M \in \mathcal{M}$:
 - (a) If $|Q_i| \geq p$ or there exists a tuple $(T, M', \rho') \in Q_i$ for some M', ρ' , return $\perp$.
 - (b) Set $\text{ctr} := 0$ and $x := \perp$. While $\text{ctr} < K$ and $x = \perp$:
 - i. Sample $\rho \xleftarrow{\$} \mathcal{R}$.
 - ii. Set $x := \text{Th}(P_i, T, M, \rho)$.
 - iii. If $\text{Prop}(x) = 1$: Insert (T, M, ρ) into Q_i .
 - iv. Else: Set $x := \perp$, $\rho := \perp$.
 - v. Set $\text{ctr} := \text{ctr} + 1$.
 - (c) If $x = \perp$: Insert $(T, M, \perp)$ into Q_i .
 - (d) Return (x, ρ) .

Output: $\mathcal{A}$ outputs (i^*, j^*, M^*, ρ^*) with $i^* \in [N]$, $j^* \in [|Q_{i^*}|]$, $M^* \in \mathcal{M}$, and $\rho^* \in \mathcal{R}$.

Winning Condition: Let $(T_{j^*}, M_{j^*}, \rho_{j^*})$ be the j^* th entry in Q_{i^*} . Output 1 if and only if:

- (i) $\text{Th}(P_{i^*}, T_{j^*}, M_{j^*}, \rho_{j^*}) = \text{Th}(P_{i^*}, T_{j^*}, M^*, \rho^*)$, and
- (ii) $(M^*, \rho^*) \neq (M_{j^*}, \rho_{j^*})$.

The advantage is:

$$\text{Adv}_{\text{Th}, N, p, \text{Prop}}^{\text{MULTI-SM-rTCR}, K}(\mathcal{A}) := \Pr[\text{MULTI-SM-rTCR}_{\text{Th}, N, p, \text{Prop}}^K(\mathcal{A}) \Rightarrow 1].$$

1.2.3 Multi-User Multi-Target Chain Preimage Resistance

Definition 1.5 (Multi-User Multi-Target Chain Preimage Resistance). Let $\text{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{H}$ be a tweakable hash function with $\mathcal{H} \subseteq \mathcal{M}$. Let $w \geq 1$ be the chain length parameter, let $N \in \mathbb{N}$ be the number of users, let $p \in \mathbb{N}$ be a per-user query bound, and let $\mathcal{A}$ be a stateful algorithm. Consider the experiment $\text{MULTI-SM-CHAIN}_{\text{Th}, N, w, p}(\mathcal{A})$:

Setup:

1. For each $i \in [N]$, sample $P_i \xleftarrow{\$} \mathcal{P}$ conditioned on $P_i \neq P_j$ for all $j < i$.
2. For each $i \in [N]$, initialize $Q_i := \emptyset$, $\text{Chains}_i := []$, and counters $\text{ctr}_i := 0$.

Query Phase: Run $\mathcal{A}$ with (classical) access to the Chain Oracle $\mathcal{O}_{\text{chain}}(i, T_1, \dots, T_{2^w-1})$ for $i \in [N]$, $(T_1, \dots, T_{2^w-1}) \in \mathcal{T}^{2^w-1}$:

- (a) If $\text{ctr}_i \geq p$, return $\perp$.
- (b) If $|\{T_1, \dots, T_{2^w-1}\}| < 2^w - 1$ or $\{T_1, \dots, T_{2^w-1}\} \cap Q_i \neq \emptyset$, return $\perp$.
- (c) Sample $Y_0 \xleftarrow{\$} \mathcal{H}$.
- (d) Compute $Y_j := \text{Th}(P_i, T_j, Y_{j-1})$ for $j = 1, \dots, 2^w - 1$.
- (e) Append $((T_1, \dots, T_{2^w-1}), (Y_0, \dots, Y_{2^w-1}))$ to Chains_i .
- (f) Set $Q_i := Q_i \cup \{T_1, \dots, T_{2^w-1}\}$ and $\text{ctr}_i := \text{ctr}_i + 1$.
- (g) Return Y_{2^w-1} .

Reveal Phase: When $\mathcal{A}$ signals to proceed:

1. Disable $\mathcal{O}_{\text{chain}}$ and provide $(P_1, \dots, P_N)$ to $\mathcal{A}$.
2. For each $i \in [N]$ and $j \in [|\text{Chains}_i|]$, initialize $\text{Revealed}_{i,j} := \emptyset$.
3. Give $\mathcal{A}$ (classical) access to the Reveal Oracle $\mathcal{O}_{\text{reveal}}(i, j, k)$ for $i \in [N]$, $j \in [|\text{Chains}_i|]$, $k \in \{0, \dots, 2^w - 2\}$:
 - (a) Let $(Y_0^{i,j}, \dots, Y_{2^w-1}^{i,j})$ be the chain values for user i , chain j .
 - (b) Set $\text{Revealed}_{i,j} := \text{Revealed}_{i,j} \cup \{k\}$.
 - (c) Return $Y_k^{i,j}$.

Output: $\mathcal{A}$ outputs (i^*, j^*, x) with $i^* \in [N]$, $j^* \in [|\text{Chains}_{i^*}|]$, and $x \in \mathcal{H}$.

Winning Condition: Let $((T_1^{i^*, j^*}, \dots, T_{2^w-1}^{i^*, j^*}), (Y_0^{i^*, j^*}, \dots, Y_{2^w-1}^{i^*, j^*}))$ be the j^* th entry in Chains_{i^*} . Let $k^* := \min(\text{Revealed}_{i^*, j^*} \cup \{2^w - 1\})$ be the earliest revealed position (or $2^w - 1$ if nothing was revealed). Output 1 if and only if:

$$\text{Th}(P_{i^*}, T_{k^*}^{i^*, j^*}, x) = Y_{k^*}^{i^*, j^*}.$$

The advantage is:

$$\text{Adv}_{\text{Th}, N, w, p}^{\text{MULTI-SM-T-CHAIN}}(\mathcal{A}) := \Pr[\text{MULTI-SM-T-CHAIN}_{\text{Th}, N, w, p}(\mathcal{A}) \Rightarrow 1].$$

Intuition 1.6. The challenger constructs secret hash chains of the form:

$$Y_0 \xrightarrow{T_1} Y_1 \xrightarrow{T_2} Y_2 \xrightarrow{T_3} \dots \xrightarrow{T_{2^w-1}} Y_{2^w-1}$$

but initially only reveals the *final* value Y_{2^w-1} . Later, the attacker may query to see some intermediate values. The attacker wins by finding a preimage of the earliest revealed value in the chain.

References

- [1] J. Drake, D. Khovratovich, M. Kudinov, and B. Wagner, “Hash-based multi-signatures for post-quantum ethereum,” Cryptology ePrint Archive, Paper 2025/055, 2025. [Online]. Available: <https://eprint.iacr.org/2025/055>